

Ahmad Tayyab

COMPUTER SCIENTIST | MACHINE LEARNING ENGINEER

ahmedtayyab84@gmail.com | +92 332 5830828 | linkedin.com/in/ahmadtayyab | github.com/ahmedtayyab

SUMMARY

Computer Science student with hands-on experience in machine learning, deep learning, and large-scale data analysis. Skilled in building end-to-end ML pipelines, feature engineering, and deploying data-driven applications.

EDUCATION

FAST-NUCES

2022 – 2026

BSc Computer Science | **Coursework:** Data Science, Deep Learning, Artificial Intelligence, Database Systems, DSA, OOP

EXPERIENCE

Full-Stack Developer (Freelance) | Self-Employed | Remote

Apr 2025 – Present

GitHub

- Developed full-stack e-commerce platform with authentication, cart system, and GST billing using PHP and MySQL.
- Deployed application on AWS EC2 using SSH and LAMP stack, improving system reliability and scalability.
- Implemented secure coding practices including password hashing, SQL injection, XSS, and CSRF protection.
- Built admin dashboard for user management and customer inquiry tracking.

PROJECTS

Supermarket Price Matching Pipeline | Python, Pandas, Web Scraping

GitHub

- Built ETL pipeline processing 500k+ records from multiple supermarket chains.
- Performed feature engineering and entity resolution using fuzzy matching; conducted price dispersion analysis.
- Detected anomalies using IQR and Z-score methods to improve data quality.

Air Quality Intelligence | Python, PCA, Streamlit, Pandas

GitHub

- Analyzed 828k+ pollution records across 100 cities using PCA (69.3% variance retained).
- Identified pollution patterns and detected 306k+ health violations through statistical analysis.
- Developed interactive Streamlit dashboard for data visualization and insights.

Titanic EDA & Visualization | Python, Pandas, Seaborn

GitHub

- Conducted exploratory data analysis with feature engineering and missing value imputation.
- Identified key survival trends based on gender and passenger class using statistical insights.

Neural Network from Scratch | NumPy, Python

GitHub

- Implemented feedforward neural network with forward and backward propagation using NumPy.
- Achieved 96% accuracy using custom ReLU and Sigmoid activation functions.

CNN Image Classification | PyTorch, Python, SVHN

GitHub

- Designed convolutional neural network with pooling layers for image classification.
- Achieved 85% accuracy on 73k+ SVHN digit images.

LSTM Time Series & Seq2Seq | PyTorch, TensorFlow

GitHub

- Built LSTM models for time series forecasting and sequence-to-sequence learning tasks.

Phobia Relief Therapy — FYP (Ongoing) | Python, AI, Sensors

GitHub

- Developing VR-based exposure therapy system using Polar H10 sensor.
- Applying decision tree models for adaptive stress response prediction.

SKILLS

Languages: Python, SQL, JavaScript, C++, C, PHP, C#

Machine Learning: Scikit-learn, Pandas, NumPy, PCA, Time Series Analysis

Deep Learning: CNNs, RNNs, LSTMs, Attention, Autoencoders

Tools & Cloud: PyTorch, TensorFlow, AWS EC2, Linux, Git, Streamlit, Jupyter